ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

# Unlock Your *Gut*, Ultimate Guide

*A Complete Functional Medicine Guide to Healing Intestinal Hyperpermeability and Ending the Chronic Inflammation Cycle*

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BEFORE YOU START

> ***Disclaimer:*** *This book is for educational purposes only and does not constitute medical advice. It does not diagnose any condition, and reading it does not make you my patient. Always consult a qualified healthcare provider before starting any new treatment, supplement, or dietary change, and do not start, stop, or change any prescribed medication without speaking to the prescriber. Before eliminating gluten, get celiac testing while you are still eating it, because removing it first costs you a clean answer permanently. Seek medical care promptly for rectal bleeding, unintentional weight loss, persistent vomiting, difficulty swallowing, fever with abdominal pain, or a change in bowel habit that does not settle.*

*The information reflects a functional medicine perspective focused on root cause identification. This is the same clinical framework I use with patients at Root Health daily.*

# Contents

INTRODUCTION

# The Gut Is Where Everything Starts

There is a pattern that repeats itself constantly in functional medicine practice. A patient arrives after seeing multiple specialists. They have been to gastroenterology, to rheumatology, to endocrinology, sometimes to psychiatry. They carry a folder of test results, all of which are reported as normal or within range. And yet they are exhausted in a way that sleep does not fix. Their thinking is slow and unreliable. Their skin breaks out in patterns no dermatologist has been able to explain. Their hormones fluctuate in ways that do not respond to the usual interventions. They gain weight despite reasonable eating. They feel anxious in a way that does not correspond to the actual circumstances of their life.

In a significant portion of these patients, the answer is in the gut. Not because the gut is the only problem, but because the gut is the entry point for a cascade of dysfunction that eventually manifests everywhere else. A compromised intestinal barrier allows substances into the bloodstream that do not belong there. The immune system mounts a response. That response, when it becomes chronic, drives systemic inflammation that affects every organ system in the body. The fatigue, the brain fog, the hormonal dysregulation, the skin conditions, the mood changes. They are downstream consequences of a gut barrier that has lost its integrity, not separate problems.

This is the clinical reality of intestinal permeability, commonly called leaky gut. It is not a fringe concept invented by alternative medicine practitioners. The tight junction proteins that regulate gut barrier function (occludin, claudin, and zonulin) have been the subject of hundreds of peer-reviewed studies. The scientist who discovered the zonulin protein, Dr. Alessio Fasano at Harvard Medical School, has published extensively on the central role of intestinal permeability in autoimmune disease, metabolic dysfunction, and neurological disorders.[1] The research is there. The clinical application of that research is what most patients have never had access to.

This guide is that access. What follows is a complete framework for understanding how the gut barrier breaks down, what specific mechanisms are responsible, how to test for it accurately, and what a systematic recovery protocol looks like: beginning with the drainage foundations that must come first, through dietary intervention, targeted supplementation, and the often-overlooked connections between gut health and thyroid autoimmunity that can change the trajectory of a patient's entire health picture.

The goal is to give you a clinical understanding of what is actually happening in your body and why this approach is sequenced the way it is, rather than a list of supplements to buy.

KEY 1

# What Leaky Gut Actually Is

The intestinal lining is one of the most remarkable structures in the human body. Stretched flat, the surface area of the small intestine alone covers an area roughly the size of a studio apartment, about 30 to 40 square meters,[2] compressed into a tube less than an inch in diameter through a system of microscopic finger-like projections called villi, and even smaller hair-like structures called microvilli that together form what is known as the brush border. This extraordinary surface area exists for one purpose: to maximize the absorption of nutrients from digested food.

But absorption cannot be indiscriminate. The same surface that absorbs nutrients must also serve as a barrier against everything in the gut lumen that should not enter systemic circulation: undigested food particles, bacterial toxins called lipopolysaccharides (LPS), fungal metabolites, parasites, and the contents of dead and lysed bacteria. The mechanism that performs this selective gatekeeping is a system of protein complexes called tight junctions.

Far from passive seals, tight junctions are dynamic structures that respond to signals from the gut environment, opening and closing in response to nutrients, pathogens, inflammatory signals, and a critical regulatory protein called zonulin. When the gut is healthy and the tight junction proteins (occludin, claudin-1, claudin-3, and ZO-1) are intact and properly assembled, the barrier functions correctly. Only properly digested nutrients cross into the bloodstream. Everything else remains in the gut lumen until it is excreted.

Intestinal permeability occurs when this tight junction system is disrupted. The gaps between intestinal cells widen. Substances that normally would not have access to the bloodstream, including partially digested proteins, bacterial endotoxins, fungal toxins, and lipopolysaccharides from the outer membranes of gram-negative bacteria, cross into systemic circulation. The immune system, which is heavily concentrated in the gut-associated lymphoid tissue (GALT) surrounding the intestinal lining, detects these foreign substances and mounts a response.

In the short term, this immune response is appropriate and protective. The problem emerges when the gut barrier remains compromised. If the tight junction disruption is not resolved, the immune activation does not resolve either. It becomes chronic. And chronic, low-grade immune activation in response to a continuously leaking gut barrier is the mechanism that drives the systemic consequences that patients experience as seemingly unrelated conditions: fatigue, brain fog, skin eruptions, hormonal instability, autoimmune activation, and metabolic dysfunction.

## Why This Is Bigger Than the Gut

The gut is far more than a digestive organ that happens to sometimes malfunction. It is arguably the most complex immunological and neurological structure in the body outside the central nervous system. Approximately 70 percent of the body's immune tissue is concentrated in the mucosa-associated lymphoid tissue of the gut. The enteric nervous system, sometimes called the second brain, contains over 100 million neurons, more than the spinal cord, and communicates bidirectionally with the brain via the vagus nerve. The gut produces more than 90 percent of the body's serotonin and significant quantities of dopamine precursors, GABA, and other neuroactive compounds.

This architecture means that what happens in the gut does not stay in the gut. When the gut barrier is chronically compromised, the consequences are systemic. LPS endotoxins crossing the gut barrier activate toll-like receptor 4 (TLR4) in immune cells throughout the body, triggering a pro-inflammatory cytokine cascade that affects the liver, the brain, the thyroid, the adrenals, and the cardiovascular system. Neuroinflammation from gut-derived immune activation has been directly linked in the research literature to depression, anxiety, cognitive impairment, and neurodegenerative disease. Thyroid dysfunction, hormonal dysregulation, and autoimmune activation are all documented downstream consequences of chronic intestinal permeability.

This is the clinical framework that explains why patients with leaky gut present with such diverse and apparently disconnected symptom pictures. The gut is not one of ten problems. In many cases, it is the upstream driver of all the others.

## The Diagnostic Gap in Conventional Medicine

Conventional gastroenterology focuses primarily on structural pathology: polyps, ulcers, tumors, masses, and the inflammatory bowel diseases that produce visible changes on colonoscopy or endoscopy. These are important conditions. But they represent only a fraction of the gut dysfunction that drives chronic illness in the broader population. A colonoscopy can be completely normal in a patient with profound intestinal permeability, significant dysbiosis, severely depleted secretory IgA, active H. pylori infection, and measurable Candida overgrowth. Every one of these findings would be invisible on a standard gastroenterological workup.

Intestinal permeability is a functional condition. It requires functional testing to identify. The zonulin antibody, the occludin/actomyosin antibodies, the tight junction protein markers, the secretory IgA measurement, and the fecal gliadin: none of these are part of routine gastroenterological or primary care practice. They have to be specifically ordered, and most physicians have not been trained in their interpretation.

This is a curriculum gap, not a criticism of conventional medicine practitioners. The gap leaves patients circulating through multiple specialties, collecting normal test results, while the actual driver of their symptoms goes undetected. The purpose of this guide is to close that gap: to give you the testing framework and the treatment roadmap that addresses what conventional testing cannot find.

KEY 2

# What Breaks the Gut Barrier

Intestinal permeability rarely develops from a single cause in isolation. In the patients who are most symptomatic and most resistant to recovery, multiple factors have converged over time to erode the gut barrier from several directions simultaneously. Understanding the specific mechanisms by which the barrier breaks down directly determines which interventions will be most effective for a given patient.

### Gliadin, Gluten, and the Zonulin Response

Of all the dietary factors that affect intestinal permeability, the relationship between gliadin, the immunogenic protein fraction of gluten found in wheat, rye, and barley, and the zonulin signaling pathway is the best documented and arguably the most clinically significant. Gliadin directly triggers the release of zonulin from intestinal epithelial cells, and zonulin is the primary physiological regulator of tight junction opening. When zonulin rises, tight junctions open. The mechanism has been demonstrated in vitro, in animal models, and in human clinical research by Dr. Fasano's group and numerous independent investigators.

What makes this particularly important is that the zonulin response to gliadin is not exclusive to patients with celiac disease. Research has demonstrated that gliadin activates the zonulin pathway in all individuals regardless of celiac status, though the magnitude and duration of the response vary substantially based on genetics, microbiome composition, and baseline gut health.[3] For patients with HLA-DQ2 or DQ8 variants, or with pre-existing gut compromise, the zonulin response to gliadin exposures can produce measurable tight junction disruption that persists for hours after consumption.

The clinical implication is significant: for patients with confirmed or suspected intestinal permeability, gluten reduction is not sufficient. Complete elimination is the therapeutic standard, at least through the active repair phase, because even small and inconsistent exposures maintain the zonulin signaling that keeps tight junctions chronically disrupted. This is why patients who "mostly" avoid gluten do not experience the same degree of improvement as those who eliminate it completely.

### Dysbiosis: When the Ecosystem Collapses

The human gut microbiome contains approximately 38 trillion microorganisms representing thousands of species.[4] The microbiome actively participates in the maintenance of gut barrier integrity through multiple mechanisms: producing short-chain fatty acids that serve as the primary fuel source for colonocytes, regulating tight junction protein expression,[5] stimulating secretory IgA production, competing with pathogens for nutrients and binding sites, and modulating the immune tone of the gut mucosa.

When this ecosystem is disrupted through antibiotic use, dietary changes, chronic stress, toxin exposure, or pathogenic infection, the consequences for gut barrier integrity can be profound. Gram-negative bacteria that overgrow in dysbiotic states produce lipopolysaccharides (LPS) as a component of their outer membrane.

When these bacteria die or are lysed, LPS is released. In a healthy gut with intact barrier function, LPS remains in the gut lumen and is excreted. In a dysbiotic gut with compromised barrier function, LPS crosses into circulation and activates the systemic inflammatory cascade through TLR4 receptors on immune cells throughout the body. This is the mechanism of what researchers call "metabolic endotoxemia," a chronically elevated circulating LPS burden that drives low-grade systemic inflammation even in the absence of any acute infection.[6]

Candida albicans presents a different but equally important mechanism. In its commensal yeast form, Candida is a normal constituent of the gut microbiome. When conditions shift toward high sugar intake, antibiotic use, immune suppression, or prolonged gut inflammation, Candida can transition to its hyphal form, in which it produces thread-like projections that physically penetrate between intestinal epithelial cells. These hyphae mechanically disrupt the tight junction architecture, creating gaps in the barrier that allow other substances to cross into circulation. Candida overgrowth also produces acetaldehyde and other metabolic toxins that damage the gut lining directly.

H. pylori, which colonizes the stomach and upper small intestine in a significant percentage of the population, affects gut barrier integrity through a different mechanism. H. pylori degrades the protective mucus layer that shields the stomach and duodenal epithelium from acid and microbial insult. Without this mucus barrier, the epithelium is exposed to chronic low-grade damage that impairs the integrity of the entire upper gut lining and predisposes to downstream intestinal permeability.

### The Role of NSAIDs, Antibiotics, and Medications

Few pharmaceuticals contribute to gut barrier disruption as consistently in clinical practice as non-steroidal anti-inflammatory drugs. NSAIDs inhibit cyclooxygenase enzymes (COX-1 and COX-2), which are responsible for producing prostaglandins throughout the body. What is often overlooked is that prostaglandins in the gut mucosa serve a protective function: they stimulate mucus secretion, regulate blood flow to the intestinal lining, and support epithelial cell turnover. When COX enzymes are inhibited, this prostaglandin-dependent mucosal protection is reduced, and the gut lining becomes more vulnerable to damage from acid, bile, and microbial products.

The gut-damaging effects of NSAIDs are well-established in the research literature. Studies using capsule endoscopy have documented mucosal erosions, ulcerations, and barrier disruption in the small intestine of patients taking NSAIDs, findings that are invisible on standard endoscopy because they occur in segments of the gut that endoscopes cannot reach.[7] Patients who rely on ibuprofen, aspirin, or naproxen regularly for pain management are often maintaining a state of chronic gut barrier compromise that prevents healing regardless of what else is done in the protocol.

Antibiotics present a different mechanism of gut disruption. Even a single course of antibiotics can produce significant alterations in gut microbial diversity that persist for six months to two years following treatment.[8] Multiple courses of antibiotics over years, which is not unusual in the history of patients with chronic illness and particularly those with recurrent sinus infections, urinary tract infections, or Lyme disease, can produce a degree of microbiome disruption that significantly impairs the ecosystem's capacity to maintain barrier integrity. The use of broad-spectrum antibiotics is not inherently problematic when clinically necessary, but the absence of microbiome restoration following antibiotic use is a significant missed opportunity in most medical settings.

### Chronic Stress and the HPA-Gut Axis

The relationship between psychological and physiological stress and gut barrier function is mechanistically direct, not metaphorical. Corticotropin-releasing hormone (CRH), released from the hypothalamus in response to stress, activates mast cells in the gut wall through CRH receptor 1. Activated mast cells release histamine, proteases, and pro-inflammatory mediators that directly increase intestinal permeability and alter gut motility. This is a rapid effect: measurable changes in gut permeability have been documented in human studies within hours of acute psychological stress exposure.[9]

Chronic stress compounds this through cortisol dysregulation. Elevated cortisol impairs tight junction protein synthesis, suppresses secretory IgA production (the gut's primary immune defense), and alters the composition of the gut microbiome in ways that favor pathogenic over commensal organisms. The clinical consequence is a gut that is perpetually operating in a defensive, pro-inflammatory state. This is why patients who are under sustained stress rarely achieve full gut barrier repair even when their diet and supplement protocols are otherwise well-designed.

The bidirectionality of this relationship is equally important to understand. A compromised gut barrier generates chronic immune activation that drives neuroinflammation via inflammatory cytokine signaling to the brain. This neuroinflammation amplifies the stress response, increases anxiety, disrupts sleep architecture, and impairs the regulatory mechanisms that would normally allow the HPA axis to return to baseline after a stressor. Gut permeability and stress dysregulation feed each other in a cycle that requires both to be addressed for either to resolve.

### Dietary Processed Foods, Seed Oils, and Emulsifiers

The industrial food supply contains several specific compounds that have been shown to impair gut barrier integrity through mechanisms independent of their general nutritional inadequacy. Emulsifiers, particularly carboxymethylcellulose (CMC) and polysorbate 80, which appear in a wide range of processed foods as texture agents, have been shown in multiple animal studies to disrupt the gut mucus layer, promote bacterial translocation across the epithelium, and drive low-grade intestinal inflammation.[10] Human data is more limited but consistent with the animal findings.

Refined seed oils high in linoleic acid, specifically canola, soybean, corn, sunflower, and safflower, present a different mechanism. The dramatic increase in dietary omega-6 fatty acids relative to omega-3 fatty acids that has accompanied the industrialization of the food supply drives the arachidonic acid cascade, generating pro-inflammatory prostaglandins and leukotrienes that degrade tight junction proteins and promote intestinal inflammation over time. The membrane phospholipids of intestinal epithelial cells incorporate the fatty acids available from the diet. A cell membrane built primarily from omega-6 fatty acids produces a fundamentally different inflammatory signaling profile than one built from omega-3s, and that signaling directly affects tight junction integrity.

Refined sugar and high-fructose corn syrup contribute to gut barrier compromise primarily through their effect on the microbiome. Pathogenic organisms, particularly Candida and certain dysbiotic bacteria, are highly preferential metabolizers of simple sugars. A diet high in refined carbohydrates selectively feeds the organisms most likely to damage the gut barrier while simultaneously depleting the fiber substrates that beneficial bacteria need to produce the short-chain fatty acids that maintain barrier integrity. The dietary change alone, removing refined sugar and processed foods, often produces measurable improvement in gut symptoms within two to three weeks, not because it directly repairs the barrier, but because it removes the primary fuel source for the organisms that are breaking it.

## Environmental Toxins and the Mold Connection

The relationship between environmental toxin burden and gut barrier integrity has become increasingly well-documented in recent years. Glyphosate, the active ingredient in Roundup herbicide and now present in detectable levels in a wide range of commercially grown foods, has been shown to inhibit tight junction proteins and alter the gut microbiome in ways that favor pathogenic organisms. Several studies have documented measurable reductions in beneficial Lactobacillus and Bifidobacterium species in glyphosate-exposed gut microbiomes, with corresponding increases in pathogenic organisms.

Mycotoxins, the toxic secondary metabolites produced by mold species in water-damaged buildings, are directly toxic to intestinal epithelial cells and have been shown to suppress secretory IgA production, impair tight junction assembly, and promote gut barrier dysfunction. This is a central clinical reason why mold illness and leaky gut so frequently coexist. The patient who comes in with a history of water-damaged building exposure and presents with gut symptoms, fatigue, brain fog, and hormonal dysregulation is not experiencing two separate problems. The mold toxin burden is a primary driver of the gut barrier dysfunction that is producing the systemic symptom picture. In these patients, gut repair protocols that do not also address the underlying mycotoxin burden produce incomplete and often temporary results.

# The Right Labs, Getting Answers That Actually Mean Something

Most patients with intestinal permeability have spent years getting tested. What they have not had is the right testing. There is a fundamental difference between the standard gastroenterological workup, which is designed to detect structural pathology, and the functional medicine approach to gut assessment, which is designed to characterize the gut's functional state: its barrier integrity, its microbial ecosystem, its immune competence, and the specific organisms and mechanisms driving dysfunction. The panels described below are what make accurate diagnosis and targeted treatment possible. But before any panel is described, one testing problem has to be stated plainly, because this book's credibility depends on it.

## The Zonulin Testing Problem, Stated Honestly

Zonulin the molecule and zonulin the lab test are in very different shape, and conflating them is where both sides of this argument go wrong.

The biology is real. Zonulin (identical to pre-haptoglobin 2) is an established regulator of tight junction opening, and the research base behind it is the legitimate science this chapter began with.[1] The commercial testing is another matter: an analysis of the most widely used commercial zonulin ELISA found that the kit does not actually detect pre-haptoglobin 2, the molecule it claims to measure, and instead recognizes other proteins, including properdin, with measured "zonulin" concentrations failing to track the haptoglobin genotypes that determine whether a person can even produce zonulin.[15] In plain terms: the number on a commercial zonulin result, serum or stool, is not a validated measurement of the molecule it is named after. **This practice does not treat any zonulin level as a diagnosis, and neither should any practice.**

Skeptics then take one step too far in the other direction. A physician on social media telling you to "skip leaky gut testing because everyone has intestinal permeability" is right about the zonulin kits and wrong about the physiology. Of course everyone has intestinal permeability; you could not absorb nutrients without it. Dismissing intestinal hyperpermeability on those grounds is like dismissing hypertension because everyone has blood pressure. The clinical question is never whether the barrier is permeable but whether it has become excessively so, and mainstream gastroenterology treats that question as real: a review in the journal Gut lays out the intestinal barrier's components, the validated research measurements of permeability (orally administered probe molecules such as the lactulose-mannitol test, and tissue-based methods), the stressed states in which barrier function measurably deteriorates, and, in the same breath, the caution that public-domain "leaky gut" claims require confirmation before endorsing exclusion diets or repair supplements.[16] Both halves of that sentence are this book's position.

What this means practically for the panels below: the antibody and stool markers they carry are read as supportive pattern evidence alongside your symptoms, your history, and your response over time, never as stand-alone verdicts, and no single number on any of them, zonulin least of all, diagnoses a leaky gut by itself.

### The Wheat Zoomer: Starting with Barrier Integrity

The Wheat Zoomer panel from Vibrant Wellness is the entry-level permeability assessment and the appropriate starting point for most patients presenting with gut-related symptoms. Its value is in what it measures directly: the antibody response to the tight junction proteins themselves. Occludin/zonulin antibodies are interpreted as evidence that the immune system has encountered these proteins, which is expected mainly when they have reached systemic circulation from a compromised epithelium. Actomyosin IgA and IgG antibodies reflect intestinal epithelial cell injury, and the actin-antibody family is the best-validated marker in this class, though that validation is strongest specifically in celiac disease with villous atrophy rather than for permeability in general. The honest grade for the panel as a whole: suggestive immunological pattern evidence, meaningfully better than a bare zonulin number, and still not a validated stand-alone diagnostic for barrier compromise. It is read alongside the clinical picture, per the testing-honesty section above, and the companion lab guide in the Gut kit walks through each marker's actual validation study by study.

The Wheat Zoomer also measures antibody reactivity to 26 different wheat and gluten peptides, providing a far more sensitive picture of gluten reactivity than the standard celiac panel. Most patients who have had standard celiac antibody testing and received a negative result have not had the breadth of gluten peptide testing that the Wheat Zoomer provides. Non-celiac gluten sensitivity, a real and measurable immunological phenomenon in which the immune system reacts to gluten peptides without producing the villous atrophy of celiac disease,[11] is identified on the Wheat Zoomer in patients who test negative on standard celiac panels. The clinical significance for gut repair is substantial: a patient cannot repair a gut barrier that gliadin is continuing to disrupt.

The Wheat Zoomer is available through rootcauseunlocked.com/order-labs and is the recommended first-line panel for patients with gut-related symptoms, food sensitivities, skin conditions, hormonal dysregulation, or brain fog without a known autoimmune diagnosis.

### The Gut Zoomer Complete: Mapping the Ecosystem

For patients with significant dysbiosis symptoms, including persistent bloating and gas, alternating bowel habits, suspected parasites, history of H. pylori, history of Lyme disease or chronic infection, or prior antibiotic use, the Gut Zoomer Complete from Vibrant Wellness provides the comprehensive microbial picture needed to make treatment precise rather than speculative. The panel characterizes over 300 gut organisms using DNA sequencing technology, identifying not just the presence of pathogenic organisms but the relative abundance and diversity of the entire microbial ecosystem.

The Gut Zoomer measures fecal zonulin (which carries the same assay-validity problem described in the testing-honesty section above, and is read here as a minor supporting data point rather than a direct permeability measure), fecal gliadin (measuring active gluten exposure in the gut lumen), secretory IgA (the gut's primary immune defense, which when depleted is consistent with chronic immune suppression and with susceptibility to ongoing pathogenic colonization), and short-chain fatty acid production, which reflects the functional capacity of the microbiome to support the colonocyte fuel supply. It identifies specific pathogenic organisms including H. pylori (with virulence factor assessment), Candida species, Giardia, Cryptosporidium, Blastocystis hominis, and a range of dysbiotic bacterial overgrowths. It provides a degree of clinical specificity that makes every subsequent treatment decision more targeted.

### The GI-MAP: A Lower-Cost Alternative

For patients who need a lower-cost alternative to the Gut Zoomer Complete, the GI-MAP from Diagnostic Solutions is a solid option. It uses quantitative PCR technology to detect specific gut pathogens with a high degree of sensitivity and is particularly useful for identifying H. pylori, Candida, parasites, and bacterial overgrowth. It is worth noting that the GI-MAP does not measure intestinal permeability markers directly, does not assess secretory IgA or short-chain fatty acid production, and does not provide the full ecosystem picture of the Gut Zoomer Complete. For patients whose primary concern is pathogen identification rather than comprehensive barrier and microbiome assessment, it is a clinically useful starting point. The GI-MAP is ordered through the office.

### The Organic Acids Test: When the Brain Is Involved

When gut symptoms are accompanied by significant brain fog, mood instability, fatigue that is disproportionate to sleep quantity, or cognitive symptoms, the Organic Acids Test (OAT) adds a dimension to the picture that gut-specific panels cannot provide. The OAT measures urinary metabolites, specifically the byproducts of cellular metabolism, microbial activity, and neurotransmitter processing, that reflect the biochemical environment in which the gut and brain are operating.

The OAT's fungal overgrowth markers, including citrinin and various Aspergillus metabolites, can identify fungal burden that is not detectable on standard stool panels because the organisms are colonizing the small intestine rather than the colon. Its neurotransmitter metabolite markers, which measure the downstream products of dopamine, serotonin, and GABA metabolism, reveal whether the neurochemical consequences of gut dysfunction are already manifesting in the nervous system. For patients whose primary complaint is fatigue and cognitive impairment alongside gut symptoms, the OAT often provides the most clinically actionable findings of any single panel. Read its markers as suggestive rather than diagnostic, the same way every other panel in this chapter is read.

### Blood Chemistry: The Systemic Inflammatory Baseline

Before or alongside any specialty gut testing, a comprehensive blood chemistry panel establishes the systemic inflammatory and metabolic baseline that contextualizes the gut findings. High-sensitivity CRP is a nonspecific screening marker: it tells you that an inflammatory load exists somewhere, not where it is coming from, so it never confirms a gut origin on its own. CBC with differential may reveal eosinophilia suggesting parasitic infection or significant food reactivity. The comprehensive metabolic panel assesses liver function, which is the primary detoxification organ bearing the burden of endotoxin clearance from a leaking gut. Thyroid markers, fasting glucose, and fasting insulin provide the hormonal and metabolic picture that determines how broadly the gut dysfunction has affected downstream systems, and fasting insulin belongs on that list alongside glucose and HbA1c because insulin rises years before glucose does. The blood chemistry panel runs through Professional Co-op.

### Testing Decision Framework

Gut symptoms + food sensitivities + no known autoimmunity: start with Wheat Zoomer

Significant bloating, altered bowels, suspected H. pylori or parasites: Gut Zoomer Complete

Lower-cost alternative to Gut Zoomer Complete: GI-MAP through the office

Brain fog, mood changes, or fatigue alongside gut symptoms: add Organic Acids Test

Confirmed or suspected Hashimoto’s: add Yersinia serology and see Key 7

All panels available at rootcauseunlocked.com/order-labs

KEY 4

# Drainage First, The Step That Makes Everything Else Work

If there is one concept in this entire guide that separates a protocol that produces lasting results from one that produces temporary improvement or makes the patient feel worse before they feel better, it is this: the drainage pathways must be open before the gut repair protocol begins. This step cannot be abbreviated or skipped. It is the clinical foundation on which every other intervention depends.

When the gut repair process is initiated, when pathogens begin dying, when the barrier starts to release accumulated debris, when the microbiome shifts and produces changes in microbial metabolite output, large quantities of cellular waste, endotoxins, and inflammatory byproducts enter the lymphatic system and systemic circulation. The body's ability to process and eliminate these substances depends on three drainage systems functioning adequately: the lymphatic system, which clears interstitial fluid and cellular waste; the liver, which conjugates and neutralizes toxins for biliary elimination; and the kidneys, which clear water-soluble metabolic waste products.

In patients with chronic gut dysfunction, all three of these drainage systems are typically under stress. The liver has been processing elevated LPS endotoxin loads for months or years. The lymphatic system is congested from chronic inflammation. Methylation, the biochemical process that drives Phase 2 liver detoxification and is essential for clearing endotoxins, hormones, neurotransmitter metabolites, and environmental toxins, is frequently impaired. When gut repair is initiated without first restoring drainage capacity, the waste products that are mobilized have nowhere to go. They recirculate, redeposit in tissues, and produce what is commonly called a Herxheimer reaction: a worsening of symptoms in the first weeks of a protocol.

**One caution about that framing, because it matters.** If feeling better proves the protocol is working and feeling worse also proves it is working, the claim cannot be tested, and an untestable claim is a poor reason to keep paying. So I treat worsening as information rather than as a milestone. Sometimes it is exactly what the paragraph above describes, and it passes. Sometimes it means the dose is too high, the drainage groundwork was skipped, or this is the wrong protocol for you. And sometimes it means something is going on that has nothing to do with the protocol at all, which is the possibility a die-off framework is worst at noticing. Significant worsening is a reason to reassess with me, not a badge. That is also why I sequence drainage before repair: not because worsening proves anything, but because the failure mode of skipping it, patients feeling dramatically worse and abandoning a protocol that might have helped them, is something I see.

## The Drainage Foundation

At Root Health, I begin every gut repair protocol with a minimum of two weeks of drainage support before any targeted repair agents are introduced. The core drainage triad consists of three products that address the three primary drainage bottlenecks.

Lymphatic and extracellular matrix support is the piece most relevant for the majority of gut dysfunction patients. The extracellular matrix, the fluid-filled connective tissue space through which cellular waste must pass on its way to the lymphatic vessels, becomes congested in patients with chronic inflammation, dietary toxin burden, and xenobiotic accumulation. A drainage botanical formula aimed at this pattern of matrix congestion, typically taken twice daily in warm water, is the appropriate choice for most patients presenting with gut dysfunction and a history of dietary toxin exposure, processed food consumption, or chemical sensitivity. Adequate water intake and daily bowel movements do more of this work than any bottle, and they cost nothing.

The drainage support has to match the stage. Acute formulas, generally built around lymphagogue botanicals such as red root, cleavers, and echinacea, suit recent inflammation and active infection. Deeper, longer-acting blends suit longstanding degeneration, heavy metal burden, and conditions present for more than three years. These are not stacked on top of each other; each addresses a distinct level of drainage compromise, and the selection depends on chronicity, severity, and the nature of the burden. For the majority of gut repair cases, the middle option aimed at matrix congestion is the right one.

Methylated B12 addresses the methylation capacity required for Phase 2 liver detoxification. The liver's ability to conjugate and neutralize the endotoxins, hormones, and environmental toxins generated during gut repair depends on an adequate supply of methyl groups. Many patients with chronic gut dysfunction have compromised methylation capacity due to nutrient depletion, genetic variants in methylation enzymes (particularly MTHFR), or the ongoing metabolic burden of processing a chronically leaking gut. A liposomal or sublingual format is preferable to an oral tablet here, because it bypasses the gut entirely, which is clinically relevant in patients whose gut is too compromised to reliably absorb what they swallow. Look for methylcobalamin rather than cyanocobalamin.

A full-spectrum electrolyte and trace mineral supplement provides the foundation required for cellular hydration and efficient drainage at the cellular level. Dehydration at the intracellular level, which can exist in patients who drink adequate water but have compromised mineral balance, is a bottleneck in drainage capacity that turns up with unusual consistency. Without adequate cellular hydration, the intracellular fluid dynamics that drive waste products from cells into the interstitial space and ultimately into lymphatic vessels are impaired. An electrolyte and trace mineral base is dosed per label direction and is maintained throughout the full repair protocol.

Two weeks on the drainage triad before introducing any gut repair agents is the minimum. In patients with significant chronicity, confirmed heavy metal burden, or a history of multiple antibiotic courses, extending the drainage phase to three or four weeks often produces substantially better outcomes in the repair phases that follow.

KEY 5

# The Anti-Inflammatory Gut Repair Diet

Dietary change is the ground on which gut repair either succeeds or fails, not a supplementary measure. The most precisely targeted supplement protocol in the world will not produce lasting gut barrier restoration in a patient who continues to consume the foods and substances that are breaking the barrier down daily. This is basic physiology, not a matter of willpower or lifestyle preference. You cannot repair a structure that is being continuously damaged.

The dietary approach in this section is organized in phases because the clinical rationale for each phase is different, and because asking a patient to make every change simultaneously increases the likelihood that none of the changes will be sustained. The sequence matters, and understanding why each phase is necessary makes compliance far more likely than a list of rules to follow.

## Phase One: The Non-Negotiables

The first phase of dietary intervention is defined by removal, specifically the removal of the dietary factors with the most direct and documented effects on gut barrier integrity. These are foods and substances that actively disrupt the tight junction architecture or feed the pathogenic organisms that are damaging it, not foods that are merely suboptimal.

Gluten and all wheat-containing foods must be completely eliminated, not reduced, during the active repair phase. The research basis is that gliadin activates the zonulin pathway in intestinal tissue regardless of celiac status,[3] so even small and intermittent gliadin exposures can re-trigger the signaling this phase is trying to quiet. Patients who attempt a “mostly gluten-free” approach do not achieve the same degree of barrier restoration as those who eliminate it completely. Cross-contamination matters. For patients who test positive for gluten reactivity on the Wheat Zoomer, eating at restaurants that use shared cooking surfaces with gluten-containing foods can be sufficient to maintain a state of chronic tight junction disruption. During the active repair phase, strict elimination is the clinical standard.

Conventional dairy, particularly casein A1 protein from most commercially available cow’s milk products, presents a mechanism similar to gliadin in genetically susceptible individuals. A1 casein is cleaved during digestion to produce beta-casomorphin-7, a peptide that has been shown to increase intestinal permeability through mechanisms that involve both direct epithelial effects and opiate receptor-mediated changes in gut motility and secretion. Some patients tolerate A2 dairy, from certain breeds of cows, goats, and sheep, without the same permeability consequences, and this can be tested during the reintroduction phase. During active repair, complete dairy elimination is the more conservative and clinically appropriate approach.

Refined sugar and high-fructose corn syrup need to be understood not as empty calories but as a specific ecological intervention in the gut microbiome. Candida and dysbiotic bacteria are highly preferential metabolizers of simple sugars. Continuing to consume refined sugar during a gut repair protocol while simultaneously attempting to reduce Candida overgrowth with supplementation is clinically counterproductive: it is the equivalent of trying to drain a bathtub while the tap is running. For patients with confirmed or suspected Candida overgrowth or significant dysbiosis, removing refined sugar is prerequisite, not optional.

Seed oils, specifically canola, soybean, corn, sunflower, safflower, and cottonseed oils, need to be replaced, not merely reduced, because their effects on gut barrier integrity operate through the composition of the cell membranes themselves. The linoleic acid in seed oils is incorporated into the phospholipid bilayers of intestinal epithelial cells, and cell membranes built primarily from omega-6 fatty acids produce a fundamentally different inflammatory signaling profile than those built from omega-3s and saturated fats. This is not a change that produces results in days, as membrane phospholipid turnover takes weeks to months, but it is a change that affects the structural integrity of every cell in the gut lining and cannot be bypassed.

Alcohol, even in moderate quantities, increases intestinal permeability through acetaldehyde, a toxic metabolite of alcohol metabolism that laboratory work indicates is particularly damaging to tight junction proteins, and through suppression of secretory IgA production. I have not found a trial establishing a safe level during an active repair protocol, and I do not assume one exists, so my clinical position is that alcohol comes out for the duration of the repair phase. In my clinical experience, the most common reason a gut repair protocol stalls after initial improvement is continued alcohol consumption, even at levels that would conventionally be described as moderate or social.

## Phase Two: What to Add

After the barrier-disrupting foods are removed, the second phase focuses on foods that actively support the repair process. This is where the nutritional quality of the diet shifts from merely avoiding harm to providing the specific substrates that the gut lining needs to regenerate.

Bone broth deserves particular attention because its therapeutic value in gut repair rests less on its mineral content than on its amino acid composition. The slow simmering of collagen-rich bones and connective tissue produces a broth concentrated in glycine, proline, and hydroxyproline: the three amino acids that are the primary structural constituents of the collagen matrix underlying the intestinal epithelium and of the tight junction proteins themselves. Glycine, in particular, has anti-inflammatory effects at the level of the gut mucosa that extend beyond its role as a collagen precursor. Daily consumption of bone broth during the active repair phase provides a sustained supply of the raw materials the gut lining needs to rebuild.

Cooked vegetables, particularly during the early repair phase, are preferable to raw for most patients with significant gut inflammation. The fiber in raw vegetables, while ultimately beneficial for microbiome diversity, requires substantial digestive work and can be irritating to an inflamed gut lining in the short term. Cooking softens the fiber matrix and makes nutrients more bioavailable. Zucchini, sweet potato, beets, carrots, winter squash, and well-cooked leafy greens provide a range of polyphenols, prebiotic fibers, and micronutrients that support both barrier repair and the reestablishment of a diverse microbiome without the irritation risk of raw fibrous vegetables.

Fermented foods warrant a clinical caveat that is frequently overlooked in generic gut health advice. Sauerkraut, kimchi, kombucha, kefir, and other fermented foods are excellent sources of diverse probiotic organisms and are genuinely supportive of microbiome recovery in patients with intact histamine clearance capacity. However, fermented foods are among the highest-histamine foods in the human diet. In patients with concurrent histamine intolerance, which is extremely common in leaky gut cases because gut permeability and DAO enzyme dysfunction frequently coexist, consuming fermented foods during the repair phase can provoke flushing, headaches, hives, heart palpitations, anxiety, and gut symptom worsening that is often misattributed to the fermented food being “detoxing” rather than recognized as a histamine reaction. The appropriate approach is to assess for histamine intolerance before introducing fermented foods rather than assuming they will be universally beneficial.

### Phase Three: Systematic Reintroduction

After 30 to 60 days of Phase One and Two implementation, tolerance to removed foods can be systematically assessed. The protocol is straightforward: introduce one food at a time, every 72 hours, and document symptom responses across the full 72-hour window. The 72-hour window is important because IgG-mediated food sensitivity reactions, the type most commonly associated with intestinal permeability, are delayed, typically appearing 24 to 72 hours after exposure rather than immediately. Patients who test single foods over 24-hour windows frequently miss the delayed reactions that are most clinically relevant.

The reintroduction process is as much a diagnostic tool as a recovery milestone. Foods that provoke reactions during reintroduction, even subtle ones such as mild brain fog, slight skin changes, increased bowel frequency, and energy changes, reveal sensitivities that had been masked by global gut inflammation during the elimination phase. These findings directly inform the long-term dietary strategy and often explain why patients with leaky gut had developed apparently paradoxical sensitivities to foods they had always previously tolerated.

KEY 6

# Strategic Supplementation for Gut Repair

The supplement protocol for gut repair is organized as a sequence, not a stack. The distinction matters clinically. A stack is a collection of supplements taken simultaneously without regard for the order of physiological events. A sequence reflects the understanding that the gut's recovery process has phases, and that introducing certain interventions before others are in place either reduces their efficacy or produces adverse reactions. The sequence described here, drainage, barrier repair, microbiome restoration, and targeted pathogen support, follows the mechanistic order in which these interventions must occur to produce the outcomes patients are looking for.

I hold no physical inventory. I keep a screened dispensary through Fullscript, which ships directly to you, and everything in the sequence below is a single ingredient or a standardized botanical you can buy from any reputable retailer instead. Where a formulation matters, it is because delivery format changes whether the compound reaches the tissue at all, and I say so in that entry rather than asking you to take it on trust. The companion supplement guide in this kit grades each of these categories against the evidence I could actually find, including what I could not find, and it is worth reading alongside this chapter.

**What you are buying here, stated plainly.** The core of this sequence has laboratory evidence, mechanistic reasoning, and long clinical use behind it, and only some of it has human outcome trials. L-glutamine has the strongest research base in the list. Colostrum, curcumin, spore probiotics, and the barrier nutrients sit further down, and the botanical antimicrobials in Phase Four sit further down still. That is not a verdict on the tools; it is what happens to compounds nobody can patent, because the trial that would settle the question never gets funded. So this sequence is reasoned clinical practice rather than trial-proven certainty, and I would rather you know which one you are following. Two consequences worth acting on: put your money into the diet and the removals in Key 5 first, because that is where the leverage and the evidence both are, and hold this protocol to an endpoint rather than running it indefinitely.

### Phase One: Drainage (Weeks 1 through 2, continued throughout)

The drainage foundation, specifically a lymphatic botanical, methylated B12, and an electrolyte and trace mineral base, is introduced in the first two weeks and maintained throughout the full protocol. As described in Key 4, these are the prerequisite interventions that determine whether everything that follows works. No other supplements are introduced until the drainage triad has been running for a minimum of two weeks.

### Phase Two: Gut Barrier Repair (Weeks 3 through 6)

L-glutamine is the primary gut barrier repair agent at Root Health and the most clinically important supplement in this phase. L-Glutamine is the preferred fuel source for intestinal enterocytes, the cells that line the gut wall, and is the primary amino acid substrate for tight junction protein synthesis. The research on L-Glutamine and gut barrier repair is extensive: it increases tight junction expression, reduces gut permeability markers, supports secretory IgA production,[12] and provides the nitrogen substrate for the rapid cell turnover that characterizes the intestinal epithelium, which replaces itself completely every three to five days. Look for a powder that combines glutamine with proline and glycine, the amino acids that support the collagen matrix underlying the epithelium. It is typically dosed at 5 to 10 g twice daily on an empty stomach, which maximizes availability to the gut epithelium before systemic tissues metabolize it.

Bovine colostrum addresses a component of gut repair that is frequently absent from conventional gut healing protocols: the restoration of gut-associated lymphoid tissue function. The GALT houses the immune infrastructure that maintains the gut's immunological relationship with its microbial inhabitants, distinguishing between commensal organisms that should be tolerated and pathogens that should be resisted. In patients with chronic gut dysfunction, prolonged NSAID use, or history of immunosuppressive medications, GALT function is often significantly impaired. Without restoring the GALT's immunological competence, the microbiome may not stabilize even after the barrier has been physically repaired, because the immune surveillance mechanism that should be regulating microbial ecology is not functioning.

A bioavailable curcumin provides upstream anti-inflammatory support at the NF-kB and COX-2 level, reducing the inflammatory signaling that impairs tight junction protein expression. Delivery format is the critical distinction here, because standard curcumin is notoriously poorly absorbed and most studies showing clinical benefit used either high doses or an enhanced delivery format such as a phytosome, nanoparticle, or piperine-paired preparation. Buying plain curcumin powder and expecting the trial results is the common mistake.

Beyond the core protocol, there is a category of gut barrier support nutrients that address the structural components of the barrier directly: compounds that fortify tight junction proteins, support the mucus layer overlying the epithelium, and provide the glycosaminoglycan substrates needed for glycocalyx integrity. These include zinc carnosine, deglycyrrhized licorice, N-acetyl glucosamine, and colostrum, among others. The companion supplement guide in this kit covers what the evidence does and does not support for each, and I have deliberately not published a fixed dosing template for them here, because the appropriate selection depends on lab findings, symptom pattern, and the presence or absence of concurrent H. pylori, NSAID history, or active gastric inflammation. That is not a hint at a secret protocol. It is the same bar this whole guide is written to: a document sold to someone I have never examined has no history, no medication list, and no way to change course next week, so what goes in it has to clear a higher bar than what I would discuss with a patient in front of me.

### Phase Three: Microbiome Restoration (Weeks 4 through 8)

A spore-based probiotic (Bacillus subtilis, Bacillus coagulans, and related species) is introduced in the fourth week, once barrier repair has been established for at least two weeks. The spore format confers a practical advantage over conventional preparations: spore-forming organisms survive gastric acid transit reliably and germinate in the small intestine, where conventional lactobacillus and bifidobacterium preparations frequently do not arrive intact. The GALT-stimulating properties of these strains are particularly relevant where secretory IgA depletion has been a significant feature of the picture.

When confirmed dysbiosis on testing indicates a more substantial microbiome reconstruction need, a high-dose multi-strain probiotic in the range of 50 billion CFU accelerates the reseeding process. This product is reserved for cases where lab findings confirm significant microbial diversity depletion. Introducing high-dose probiotics into a gut with significant ongoing inflammation and an unrepaired barrier can paradoxically worsen symptoms by increasing the microbial antigen load before the immune system is prepared to handle it.

Saccharomyces boulardii serves a specific and important role that is distinct from bacterial probiotic preparations. S. boulardii is a beneficial yeast that does not permanently colonize the gut but produces transient clinical effects that are highly valuable during the repair phase: it competes directly with Candida for mucosal binding sites, it stimulates secretory IgA production, it produces proteases that degrade Clostridium difficile toxins, and it stabilizes the gut barrier through effects on tight junction protein expression. For any patient with confirmed or suspected Candida overgrowth, which should be considered in any patient with a history of antibiotic use, sugar-heavy diet, or positive OAT fungal markers, S. boulardii is a standard component of the microbiome restoration phase.

### Phase Four: Targeted Pathogen Support (Weeks 6 through 12)

Targeted antimicrobial and pathogen-specific interventions are the last category to be introduced, not the first. This sequence is counterintuitive for many patients, who assume that identifying a pathogen on a lab result means addressing the pathogen is the highest priority. The clinical reality is that antimicrobial interventions in a gut with unrepaired barrier function and insufficient drainage capacity are tolerated poorly, and often produce a rebound in the very pathogen populations they are targeting, because the competitive microbial community that would normally keep them in check has not been restored. The caution stated in Key 4 applies here in full: feeling worse in this phase is information to act on, not proof the protocol is working.

Once drainage is established and the barrier has been supported for at least two weeks, pathogen-specific interventions are selected based on lab findings. For Candida overgrowth, pau d'arco and enteric-coated oregano oil standardized for carvacrol form the botanical antifungal core, and the full sequencing is covered in the companion Candida material. For H. pylori, mastic gum, and for general bacterial overgrowth and dysbiosis, berberine and a broad-spectrum botanical antimicrobial, are the usual choices. For Yersinia enterocolitica, the foodborne pathogen with documented thyroid autoimmune consequences, berberine has shown in vitro activity, and it is combined with gut barrier repair to reduce the antigen traffic driving molecular mimicry. Note that in vitro activity is a laboratory finding rather than a demonstrated clinical outcome, and that confirmed H. pylori is a condition where prescription eradication therapy has the evidence behind it.

| Phase | Products | Duration | Clinical Goal |
|---|---|---|---|
| 1. Drainage | Lymphatic botanical blend, methylated B12, electrolyte and trace mineral base | Weeks 1-2 min; continue throughout | Open lymphatic, liver Phase 2, cellular hydration |
| 2. Barrier Repair | L-glutamine with glycine and proline, colostrum, bioavailable curcumin, plus targeted barrier nutrients such as zinc carnosine and deglycyrrhizinated licorice | Weeks 3-6 | Rebuild tight junctions, restore mucosal immunity |
| 3. Microbiome | Spore-based probiotic, high-dose multi-strain probiotic if indicated, Saccharomyces boulardii | Weeks 4-8 | Reseed diversity, restore secretory IgA, displace Candida |
| 4. Pathogen | Selected by lab findings: pau d'arco, enteric-coated oregano oil, mastic gum, berberine | Weeks 6-12 | Targeted elimination with drainage and barrier in place |

> ***Where to get these***
>
> *Every item in this table is a single ingredient or a standardized botanical available from any reputable retailer. Look for independent third-party testing and match the form named here rather than the name on the front of the bottle. If you would rather not evaluate that yourself, I keep a screened dispensary through Fullscript, which ships directly to you, organized by these phases.*
>
> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. Nothing in this book was included, excluded, ranked, or dosed based on what it earns, and you can buy any of it anywhere.*

KEY 7

# When the Gut Attacks the Thyroid

Leaky gut does not stay local. In some patients, a compromised gut barrier goes beyond contributing to systemic symptoms. It is the hidden driver of a specific autoimmune process targeting a completely different organ: the thyroid gland. Two infectious organisms that originate in the gut have well-documented mechanisms by which they trigger or amplify Hashimoto's thyroiditis. Understanding these connections may be the most important clinical insight in this entire guide for patients who have both gut symptoms and thyroid problems that are not responding to conventional treatment.

## Yersinia Enterocolitica and the Thyroid

Yersinia enterocolitica is a gram-negative foodborne bacterium transmitted primarily through contaminated pork products, unpasteurized dairy, and undercooked meats. In immunocompetent individuals, acute Yersinia infection produces a self-limiting gastroenteritis that resolves within a few weeks. What is not widely appreciated is that Yersinia can persist in the gut in a subclinical state in susceptible individuals, and in doing so it triggers a molecular mechanism with thyroid consequences that conventional medicine rarely considers.

Yersinia produces an outer membrane protein that shares structural homology with the human TSH receptor on thyroid cells. When the immune system generates antibodies against this Yersinia surface protein, which is appropriate and necessary, those antibodies cross-react with thyroid tissue through a mechanism called molecular mimicry.[13] The immune system, unable to distinguish between the bacterial epitope and the thyroid receptor it resembles, mounts an autoimmune attack on the thyroid gland. The result is the elevated TPO and thyroglobulin antibodies characteristic of Hashimoto's thyroiditis. Multiple studies have documented significantly elevated Yersinia antibodies in Hashimoto's patients compared to healthy controls.[14] Worth stating precisely: that is an association between exposure and thyroid autoimmunity, and the mimicry mechanism proposed to explain it is still argued in the literature rather than settled. What is not in doubt is that the association is consistent enough to be worth looking for in a patient whose thyroid antibodies will not settle.

The clinical implication is important: Yersinia is a gut pathogen with a thyroid consequence. It is not covered on standard thyroid panels. It is not a tick-borne organism and is not covered on tick-borne disease panels. It requires a standalone order: LabCorp Yersinia enterocolitica IgG, IgA, and IgM serology. Treating Hashimoto's without assessing for Yersinia is treating the downstream autoimmune consequence while leaving a primary upstream trigger unaddressed.

## Lyme Disease and the Thyroid

Borrelia burgdorferi, the primary causative organism in Lyme disease, has an independent and clinically significant relationship with Hashimoto's thyroiditis that goes beyond the general immunological dysfunction of chronic Lyme. Borrelia produces surface proteins with structural similarity to thyroid antigens, triggering cross-reactive immune responses through the same molecular mimicry mechanism as Yersinia.[13] Chronic Lyme disease also drives a sustained Th17-dominant immune environment, the same immune polarization that characterizes the lymphocytic thyroid infiltration of Hashimoto's, making the two conditions mutually reinforcing in patients who have both.

The gut is the shared mechanism that ties these two conditions together. Lyme disease suppresses secretory IgA production, promotes gut dysbiosis, and compromises gut barrier integrity. A leaky gut in a patient with active or past Lyme infection creates the persistent antigen trafficking that keeps both the Lyme immune response and the thyroid autoimmunity running simultaneously. In clinical practice, patients with Hashimoto's whose antibody levels do not normalize despite appropriate dietary intervention and gut repair should be evaluated for tick-borne co-infection, because the immune dysregulation driving their antibody elevation may have an infectious origin that gut repair alone cannot resolve.

### The Key Point for Hashimoto's Patients

Both Yersinia and Lyme are gut-mediated triggers for Hashimoto's thyroiditis. Treating the gut is a prerequisite for thyroid autoimmune recovery, but in some patients, the gut alone is insufficient. The full testing and treatment protocols for both connections, including the Yersinia serology sequence, the Lyme co-infection workup, and the complete Hashimoto's supplement protocol, are in the companion guide:

**Unlock Your Hashimoto's** · rootcauseunlocked.com/hashimotos

KEY 8

# Your Next Steps with Root Health

The framework in this guide gives you something that most patients with gut-related chronic illness have never had: a mechanistic understanding of what is actually happening in their body, why the standard workup misses it, and what a systematic approach to recovery actually looks like. That understanding is clinically valuable on its own. But it reaches its full potential when it is applied to your specific case, with your labs, your history, and your specific pattern of dysfunction.

Gut dysfunction is not a one-size-fits-all condition. The specific organisms driving your dysbiosis, the degree of your barrier compromise, the genetic variants affecting your detoxification capacity, the co-conditions, including thyroid autoimmunity, tick-borne infection, mold burden, and hormonal dysregulation, all of which interact with your gut picture all require clinical interpretation. The protocols in this guide are the clinical framework. Individualized care is what makes them work for you specifically.

## Start with the Right Testing

Your entry lab panel should match your clinical picture. For patients with gut symptoms, food sensitivities, and no known autoimmunity, the Wheat Zoomer is the right starting point. It reads on barrier integrity and gluten reactivity, as suggestive pattern evidence rather than as a stand-alone diagnosis. For patients with significant bloating, altered bowels, or suspected pathogenic overgrowth, the Gut Zoomer Complete provides the most comprehensive microbial picture. If cost is a primary consideration, the GI-MAP, available through the office, is a lower-cost alternative. When brain fog, anxiety, or fatigue are prominent alongside gut symptoms, the Organic Acids Test adds the neurochemical dimension. For patients with confirmed or suspected Hashimoto's, Yersinia serology belongs in the workup. All panels are available at rootcauseunlocked.com/order-labs.

Before you order any of them, read the testing-honesty section in Key 3 again. None of these panels confirms a leaky gut by itself, and a panel ordered without a clear answer to “if this comes back positive, what specifically changes about my care” is money spent on curiosity rather than on a decision.

## Book a Root Discovery Session

The Root Discovery Session is a free clinical conversation with me. It is not a sales call. It is a focused clinical intake where your symptom history, prior testing, and current clinical picture are reviewed to identify the most targeted path forward for your case. If you are uncertain which lab panel is right for you, or if you want clinical input on interpreting prior results before investing in additional testing, this is where that conversation happens. Book at rootcauseunlocked.com/discovery, or call the Garner, NC office at 919-662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

## If you would rather start on your own

If you would rather start on your own, the highest-value work in this guide costs nothing to begin. Take the Key 5 removals seriously for eight weeks, get celiac testing before you drop gluten rather than after, keep a written baseline so you can tell whether anything actually moved, and give every restriction an endpoint. Track it at rootcauseunlocked.com/tracker. A month of honest data makes any future appointment, with me or with your own physician, far more useful than memory alone.

## A Personal Note

If you have been suffering with gut symptoms, fatigue, brain fog, skin conditions, or hormonal disruption that nobody has been able to explain: please know that there is almost always a root cause that has not yet been found. A normal colonoscopy is not a complete gut workup. Normal bloodwork is not normal function. You deserve the right tests with the right clinical interpretation, and a protocol designed around your specific case. That is exactly what Root Health is built for.

**Dr. Adam Seay, DC** | Root Health, Garner NC | rootcauseunlocked.com

Root Health | Garner, NC | rootcauseunlocked.com/gut

# References


1. Fasano A. Zonulin and its regulation of intestinal barrier function: the biological door to inflammation, autoimmunity, and cancer. Physiological Reviews. 2011. https://pubmed.ncbi.nlm.nih.gov/21248165/

2. Helander HF, Fändriks L. Surface area of the digestive tract, revisited. Scandinavian Journal of Gastroenterology. 2014. https://pubmed.ncbi.nlm.nih.gov/24694282/

3. Drago S, El Asmar R, Di Pierro M, Grazia Clemente M, Tripathi A, Sapone A, Thakar M, Iacono G, Carroccio A, D'Agate C, Not T, Zampini L, Catassi C, Fasano A. Gliadin, zonulin and gut permeability: Effects on celiac and non-celiac intestinal mucosa and intestinal cell lines. Scandinavian Journal of Gastroenterology. 2006. https://pubmed.ncbi.nlm.nih.gov/16635908/

4. Sender R, Fuchs S, Milo R. Revised Estimates for the Number of Human and Bacteria Cells in the Body. PLoS Biology. 2016. https://pubmed.ncbi.nlm.nih.gov/27541692/

5. Peng L, Li ZR, Green RS, Holzman IR, Lin J. Butyrate enhances the intestinal barrier by facilitating tight junction assembly via activation of AMP-activated protein kinase in Caco-2 cell monolayers. The Journal of Nutrition. 2009. https://pubmed.ncbi.nlm.nih.gov/19625695/

6. Cani PD, Amar J, Iglesias MA, Poggi M, Knauf C, Bastelica D, Neyrinck AM, Fava F, Tuohy KM, Chabo C, Waget A, Delmée E, Cousin B, Sulpice T, Chamontin B, Ferrières J, Tanti JF, Gibson GR, Casteilla L, Delzenne NM, Alessi MC, Burcelin R. Metabolic endotoxemia initiates obesity and insulin resistance. Diabetes. 2007. https://pubmed.ncbi.nlm.nih.gov/17456850/

7. Watanabe T, Fujiwara Y, Chan FKL. Current knowledge on non-steroidal anti-inflammatory drug-induced small-bowel damage: a comprehensive review. Journal of Gastroenterology. 2020. https://pubmed.ncbi.nlm.nih.gov/31865463/

8. Jernberg C, Löfmark S, Edlund C, Jansson JK. Long-term ecological impacts of antibiotic administration on the human intestinal microbiota. The ISME Journal. 2007. https://pubmed.ncbi.nlm.nih.gov/18043614/

9. Vanuytsel T, van Wanrooy S, Vanheel H, Vanormelingen C, Verschueren S, Houben E, Salim Rasoel S, Tóth J, Holvoet L, Farré R, Van Oudenhove L, Boeckxstaens G, Verbeke K, Tack J. Psychological stress and corticotropin-releasing hormone increase intestinal permeability in humans by a mast cell-dependent mechanism. Gut. 2014. https://pubmed.ncbi.nlm.nih.gov/24153250/

10. Chassaing B, Koren O, Goodrich JK, Poole AC, Srinivasan S, Ley RE, Gewirtz AT. Dietary emulsifiers impact the mouse gut microbiota promoting colitis and metabolic syndrome. Nature. 2015. https://pubmed.ncbi.nlm.nih.gov/25731162/

11. Sapone A, Bai JC, Ciacci C, Dolinsek J, Green PH, Hadjivassiliou M, Kaukinen K, Rostami K, Sanders DS, Schumann M, Ullrich R, Villalta D, Volta U, Catassi C, Fasano A. Spectrum of gluten-related disorders: consensus on new nomenclature and classification. BMC Medicine. 2012. https://pubmed.ncbi.nlm.nih.gov/22313950/

12. Achamrah N, Déchelotte P, Coëffier M. Glutamine and the regulation of intestinal permeability: from bench to bedside. Current Opinion in Clinical Nutrition and Metabolic Care. 2017. https://pubmed.ncbi.nlm.nih.gov/27749689/

13. Benvenga S, Guarneri F, Vaccaro M, Santarpia L, Trimarchi F. Homologies between proteins of Borrelia burgdorferi and thyroid autoantigens. Thyroid. 2004. https://pubmed.ncbi.nlm.nih.gov/15671776/

14. Chatzipanagiotou S, Legakis JN, Boufidou F, Petroyianni V, Nicolaou C. Prevalence of Yersinia plasmid-encoded outer protein (Yop) class-specific antibodies in patients with Hashimoto's thyroiditis. Clinical Microbiology and Infection. 2001. https://pubmed.ncbi.nlm.nih.gov/11318812/

15. Scheffler L, Crane A, Heyne H, Tönjes A, et al. Widely Used Commercial ELISA Does Not Detect Precursor of Haptoglobin2, but Recognizes Properdin as a Potential Second Member of the Zonulin Family. Frontiers in Endocrinology. 2018;9:22. https://pubmed.ncbi.nlm.nih.gov/29459849/

16. Camilleri M. Leaky gut: mechanisms, measurement and clinical implications in humans. Gut. 2019;68(8):1516-1526. https://pubmed.ncbi.nlm.nih.gov/31076401/



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# From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

# Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

**Seek medical care now** for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain, severe or worsening abdominal pain, or anemia. Those are not leaky gut. They are reasons for a gastroenterologist, and attributing them to intestinal permeability is how a real diagnosis gets missed.

# Key 2 applied: what actually breaks the barrier

*Unlock Your Gut* devotes its second key to what damages the barrier in the first place, and that list is the spine of this whole kit. A repair protocol that never removes the thing doing the damage is a repair protocol you will be repeating.

| What breaks it | Why it matters | What to do about it |
|---|---|---|
| **Medications** | NSAIDs, repeated antibiotics, and long-term acid suppression all act directly on the barrier or the community living behind it | Review the list with your prescriber. Some are necessary and some are habit |
| **Dysbiosis** | The bacterial community is part of the barrier, not a passenger behind it | Stool analysis where symptoms justify it, then rebuild rather than only kill |
| **Diet** | Low fiber diversity, high ultra-processed intake, and alcohol | Covered in the diet guide, which is the longest lever here |
| **Stress** | Not a metaphor. Measured directly in humans, see below | Daily down-regulation, treated as part of the protocol |
| **Sleep** | Repair happens overnight, and the gut-brain signalling runs both ways | A fixed sleep window, protected like a medication |
| **Infection** | An acute gut infection is one of the best-documented ways this starts | History matters. Post-infectious onset changes expectations |

**Notice how many of those are not supplements.** That is the honest shape of gut repair, and it is why the supplement guide is not the first document in this kit.

# The honest starting position

**Intestinal permeability is real.** The barrier exists, it can be measured, it is affected by medications, alcohol, and specific dietary components, and it is altered in several diseases. None of that is fringe.

**The market that grew around it is a different matter.** Several tests are sold as confirming leaky gut. Their validation ranges from genuinely good to essentially absent, and the ones sold hardest are not always the ones with the most behind them.

This guide sorts them. That is more useful to you than another explanation of tight junctions.

# The markers, ranked by what actually backs them

**Read your panel as a pattern, not as a verdict.** These markers measure different things, and treating them as several confirmations of one fact overstates what you have.

## IgA anti-actin, the best validated on the panel

In a multicentre study, IgA anti-actin antibodies showed **80% sensitivity and 85% specificity** for celiac disease with villous atrophy, with an area under the curve of 0.873.[1] Anti-actin levels tracked with the severity of intestinal damage.

**The honest caveat:** that validation is specifically for celiac disease with villous atrophy, not for permeability in general. In a broader celiac population a separate study found much weaker performance, 41.3% sensitivity and 71.4% specificity, and detection was lower in people already following a gluten-free diet.[2]

**So it is a good marker of intestinal damage severity in a specific disease, and a weaker one outside it.** That is still the strongest thing on the panel.

## LPS and endotoxin antibodies, measuring something different

These reflect bacterial products reaching systemic circulation. In liver disease across all stages, gut-vascular barrier and bacterial translocation markers including endotoxin-core IgA antibodies correlated with each other **but not with intestinal damage markers**.[3] In a cystic fibrosis study, LPS antibody showed reliable intraindividual stability.[4]

**Why this matters for reading your report.** Translocation and mucosal damage are different processes. An elevated LPS antibody alongside a normal damage marker is not a contradiction to explain away, it is two different findings.

## Serum zonulin, where the caution has hardened into a verdict

**The assay problem is worse than a failed correlation: the most widely used commercial zonulin ELISA was shown not to detect zonulin at all.** An analysis of the kit found it does not detect pre-haptoglobin 2, the molecule zonulin actually is, and instead recognizes other proteins, including properdin, with measured concentrations failing to track the haptoglobin genotypes that determine whether a person can even produce zonulin.[11] Separately, serum zonulin by commercial kit failed to correlate with the physiologic lactulose-mannitol measure of permeability ($r^2 = 0.004$ in 39 first-degree relatives of Crohn's patients), or with intestinal fatty acid binding protein.[5]

**The position this kit takes, plainly: skip zonulin testing, serum or stool, until a validated assay exists.** The zonulin molecule's biology is real science. The number sold under its name is not a validated measurement of it, and this practice will not interpret one as a diagnosis. If a panel you already ran includes a zonulin value, treat it as uninterpretable rather than as evidence in either direction. Note also that serum zonulin protein and **anti-zonulin antibodies** are different measurements; panels vary in which they run, and the assay critique above concerns the former.

**And the equally important other half, because a broken test does not un-discover the physiology.** Intestinal hyperpermeability is real, mainstream gastroenterology: a review in the journal Gut details the barrier's components, the validated research measures of permeability (probe-molecule tests like lactulose-mannitol, and tissue methods), and the stressed states in which barrier function measurably deteriorates, while cautioning that public-domain "leaky gut" claims need confirmation before exclusion diets or repair supplements are endorsed.[12] Everyone has intestinal permeability; you could not absorb nutrients without it. The clinical question is excess, exactly as the clinical question in hypertension is not whether you have blood pressure. Anyone who dismisses the entire territory because a bad commercial test exists has confused the assay with the physiology.

## Occludin antibodies

I searched for human validation of occludin antibodies as a barrier marker and did not find a comparable body of work. That is not a claim they are useless. It is a statement about what I could and could not verify, and you should weight the result accordingly.

## The lactulose-mannitol test, the physiologic standard with a practical problem

This is the closest thing to a direct functional measure: you drink two sugars and their recovery ratio in urine reflects barrier function.

**Its problem is variability.** In a study running 6,602 tests across 8 countries, mean mannitol recovery ranged from 1.34% to 5.88% between sites, lactulose recovery from 0.19% to 0.58%, with strong sex differences and age effects.[6] That work was in young children in diverse global settings, so the exact figures do not transfer to an adult in North Carolina, but the lesson does: **this test needs appropriate reference norms, and a single number without them is hard to interpret.**

## The commercially available version, read honestly

A dual-sugar profile is commercially available as Genova Diagnostics' Intestinal Permeability Assessment, and its own sample report documentation supports an honest evaluation. It reports three values: lactulose percent recovery (reference at or below 1.50 percent), mannitol percent recovery (reference 4 to 27 percent), and the lactulose/mannitol ratio (reference at or below 0.10), calculated from pre-drink and post-drink urine levels, which corrects each result against your own baseline. The interpretation frame matches the physiology described above: elevated lactulose points at paracellular leak, elevated mannitol at transcellular permeability, low mannitol at malabsorption, and the ratio carries the headline signal, since it cancels the transit, hydration, and kidney factors that affect both sugars together. The report's own sample case shows why the ratio matters: both individual sugars within range while the ratio runs elevated.

Three transparency points, all from the vendor's own documentation, cited by name. The stated methodology is enzymatic rather than mass spectrometry, which is workable but is the softer quantification method, and it argues for reading single results conservatively. The reference ranges are population-statistical intervals (95 percent of a reference population), with Genova's own printed caution that values between one and two standard deviations are not necessarily abnormal and need clinical correlation. And like essentially every test in this category, it is a lab-developed test that has not been FDA-cleared, which the report states plainly.

**The strongest use of this test in our model is baseline and retest, not one-time diagnosis.** A single result inherits every caveat above. A change, measured by the same lab with the same assay and the same reference frame, before and after a defined eight-to-twelve-week intervention window, is the most defensible objective permeability evidence currently available to an outpatient, and it is the one thing no antibody panel or zonulin number can honestly provide. If you run it, run it twice, same lab both times, and let the delta, not the single number, carry the conclusion.

## Multi-marker panels, where the evidence is better than for single markers

Combining markers does carry real information in defined clinical contexts. Serum biomarkers of intestinal barrier dysfunction have been studied for predicting postoperative endoscopic recurrence in Crohn's disease.[7]

**That is the strongest argument for a panel**, and also the clearest statement of its limits: the validation exists inside specific diseases with specific questions, not as a general leaky gut score.

# What actually damages the barrier, which is more useful than measuring it

A review of human intestinal barrier function is unusually direct about this. **Stressors generally reduce barrier function**, and the named ones include nonsteroidal anti-inflammatory drugs, exercise, and pregnancy. Dietary factors that **increase** permeability include ethanol, fructose, and dietary emulsifiers. Factors reported to improve the barrier include vitamins A and D, tryptophan, cysteine, and fiber.[8]

**Why I am putting this in the lab guide.** For most people the actionable finding is not a number. It is that they take ibuprofen several times a week, drink most nights, and eat a diet built on emulsified processed food. Those are established barrier stressors and none of them requires a test to identify.

# What to rule out before you conclude leaky gut

Permeability is a mechanism, not a diagnosis. These are diagnoses, they are treatable, and they get missed when everything is attributed to the barrier.

| Worth ruling out | Why |
|---|---|
| Celiac disease | Real diagnosis, real treatment. **Test while still eating gluten** |
| Inflammatory bowel disease | Fecal calprotectin is the usual screen; blood in stool is never leaky gut |
| Small intestinal bacterial overgrowth | Bloating, distension, and irregularity overlap almost completely |
| Colorectal cancer | Which is why weight loss, bleeding, and anemia are red flags, not symptoms to interpret |
| Pancreatic insufficiency | Fatty stool, weight loss, and it is directly testable |
| Microscopic colitis | Watery diarrhea with a normal-looking colonoscopy unless biopsied |
| Bile acid diarrhea | Common, treatable, and routinely mistaken for IBS |
| Thyroid disease | Motility changes both directions |

# What to ask your clinician

- “Have I been screened for celiac disease, and do I need to be eating gluten for it to be valid?”
- “Would a fecal calprotectin be reasonable to rule out inflammatory bowel disease?”
- “I take NSAIDs regularly. Could that be contributing, and is there an alternative?”
- “If this permeability panel comes back positive, what would you change about my care?” The best question on the list, because if the answer is nothing, the test is not worth running.

**Bring:** a symptom timeline, your full medication list including over-the-counter NSAIDs, honest alcohol intake, and any previous results including ones you paid for yourself.

# Reading your results as a pattern

**Celiac serology positive.** Stop here and pursue that. It is a real diagnosis with a treatment that works, and it outranks everything else on the page.

**Calprotectin elevated.** Needs gastroenterology. Not a supplement problem.

**Barrier markers elevated, everything else negative, symptoms real.** Reasonable to work on the modifiable stressors above. Not a reason for an open-ended protocol.

**Everything negative, symptoms real.** Common. It means the barrier hypothesis has not been supported, not that you are imagining anything, and the search should continue.

# When to seek clinical review

**Urgently:** blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain, or severe abdominal pain.

**Promptly:** new symptoms after starting a medication, or a change in bowel habit lasting more than a few weeks, particularly over age 45.

**Before eliminating gluten:** get celiac testing first, while still eating it.

# Track what changes

**Root Tracker:** rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=gut-lab

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# Working with Root Health

**Book a free Root Discovery Session:** rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=gut-lab

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

The question list above also works in any office.

**Get in touch:** rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=gut-lab

# Stress and sleep, which are barrier inputs rather than lifestyle advice

The reason these get their own section is that in the gut, unlike in most conditions, the stress effect has been measured directly rather than inferred.

**Stress changes permeability, measurably, in healthy people.** In an experimental study, acute stress produced a marked increase in intestinal macromolecular permeability, measured as albumin permeability, in healthy women but not in men. The authors had previously shown that chronic psychosocial stress induces jejunal epithelial barrier dysfunction, and they propose this mechanism as a contributor to female oversusceptibility to irritable bowel syndrome (PMID 22625665).

Read that precisely, because the detail matters and most summaries drop it: the effect was **significant in women and not in men** in this study. That is a real finding rather than a rounding error, and if you are a woman with a stress-reactive gut, it is the closest thing to direct evidence you will find that the connection is physical.

**Sleep sits on the same axis.** A 2026 review of sleep disruption and the gut-brain dialogue describes bidirectional communication between the microbiota and the brain and the way sleep loss interacts with microbial composition, while being explicit that mechanistic and translational work is still needed (PMID 42092445). Directional, not settled, and pointing the same way as everything else here.

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- Five to ten minutes daily of slow breathing, longer exhale than inhale. Consistency beats technique.
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- Eat sitting down and unhurried. Digestion is parasympathetic, and a rushed meal is a worse meal regardless of what is on the plate.
- Movement most days at a sustainable intensity.

None of that costs anything, none of it interacts with anything, and leaving it out is why some people run three gut protocols and stay where they started.

# References


1. Carroccio A, et al. IgA anti-actin antibodies ELISA in coeliac disease: a multicentre study. *Dig Liver Dis.* 2007. PMID 17652043.
2. IgA anti-actin antibodies in celiac disease. *Gastroenterol Clin Biol.* 2011. PMID 20655155.
3. Surrogate Markers of Intestinal Permeability, Bacterial Translocation and Gut-Vascular Barrier in Liver Disease. *Liver Int.* 2025. PMID 40317887.
4. Quantification of Enteric Dysfunction in Cystic Fibrosis: Inter- and Intraindividual Variability. *J Pediatr.* 2024. PMID 37866678.
5. Serum Zonulin Measured by Commercial Kit Fails to Correlate With Physiologic Measures of Altered Gut Permeability in First Degree Relatives of Crohn's Disease Patients. *Front Physiol.* 2021. PMID 33841181. *n=39, healthy population, one commercial kit.*
6. Age and Sex Normalization of Intestinal Permeability Measures: Results From the MAL-ED Study. *J Pediatr Gastroenterol Nutr.* 2018. PMID 28644347. *Pediatric, multi-country.*
7. Serum biomarkers of intestinal barrier dysfunction predict postoperative endoscopic recurrence in Crohn's disease. *BMC Gastroenterol.* 2026. PMID 41023842.
8. Camilleri M, Vella A. Human Intestinal Barrier: Effects of Stressors, Diet, Prebiotics, and Probiotics. *Clin Transl Gastroenterol.* 2021. PMID 33492118.
9. Alonso C, Guilarte M, Vicario M, et al. Acute experimental stress evokes a differential gender-determined increase in human intestinal macromolecular permeability. *Neurogastroenterol Motil.* 2012;24(8):740-746. PMID 22625665
10. Andersen ML, Alvarenga TA, Tufik S. Cognition in the time of dysbiosis: Sleep disruption and the gut-brain dialogue. *Prog Neuropsychopharmacol Biol Psychiatry.* 2026;147:111730. PMID 42092445
11. Scheffler L, Crane A, Heyne H, et al. Widely Used Commercial ELISA Does Not Detect Precursor of Haptoglobin2, but Recognizes Properdin as a Potential Second Member of the Zonulin Family. *Front Endocrinol (Lausanne).* 2018;9:22. PMID 29459849
12. Camilleri M. Leaky gut: mechanisms, measurement and clinical implications in humans. *Gut.* 2019;68(8):1516-1526. PMID 31076401




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# Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

**Seek care now** for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, or fever with abdominal pain.

**Get celiac testing before eliminating gluten**, while you are still eating it. This comes up in every guide in this kit because it is the single most common self-inflicted mistake in gut health, and it costs you a clean answer permanently.

# Remove before you add

*Unlock Your Gut* Key 2 lists what breaks the barrier in the first place: medications, dysbiosis, diet, stress, sleep, and infection. The lab guide turns that list into a table you can check yourself against. Work the ones that are yours, because a repair protocol that leaves the damage running is one you will repeat.

This is the organizing idea of the guide, and it is the opposite of how gut protocols are usually sold.

A review of human intestinal barrier function names the dietary factors that **increase** permeability: **ethanol, fructose, and dietary emulsifiers**. It names non-steroidal anti-inflammatory drugs among the stressors that reduce barrier function. And it lists factors that improve the barrier: vitamins A and D, tryptophan, cysteine, and fiber.[1]

Notice the asymmetry. The things that damage the barrier are specific, common, and mostly under your direct control. The things that improve it are ordinary nutrition. There is no exotic protocol in that list, on either side.

**So the sequence is: stop the four documented stressors, feed the barrier ordinary food, and only then consider whether anything else is needed.** That order is free, it is evidence-aligned, and it is the reverse of buying a repair supplement while the cause continues.

# The four stressors, in order of how much control you have

## Alcohol

Ethanol is named directly as increasing permeability.[1] This is not a moralizing point and I am not going to give you a safe number, because I do not have one for a person with gut symptoms.

**What to do:** a genuine four-week break, not a reduction, is one of the highest-yield experiments in this guide. If symptoms improve, you have learned something no panel would have told you.

## NSAIDs

Ibuprofen, naproxen, and the rest are named among the stressors that reduce barrier function.[1] A great many people with gut symptoms take them several times a week for headaches, periods, or joint pain and never mention it, because it does not feel like medication.

**What to do:** count your actual weekly use honestly. If it is regular, that is a conversation with your prescriber about alternatives, not something to stop abruptly if you take them for a diagnosed condition.

## Dietary emulsifiers

These are the additives that keep processed food smooth and stable: carboxymethyl cellulose, polysorbate 80, carrageenan, lecithins.

**There is now human trial data.** A placebo-controlled randomised trial put 60 healthy participants on an emulsifier-free diet for two weeks, then randomised them to four weeks of carboxymethyl cellulose, polysorbate 80, carrageenan, soy lecithin, native rice starch, or no additive, delivered in brownies.[2] After the two-week emulsifier-free run-in, cholesterol fell. Under supplementation, overall microbial diversity stayed stable but composition shifted by treatment, and short-chain fatty acid concentrations were lower with carrageenan.

**Read that at the right size.** It is 60 healthy people, four weeks, measuring mechanism rather than disease. It is not proof that emulsifiers cause illness. It is real human evidence that specific common additives change the gut environment measurably, in a design good enough to take seriously.

**What to do:** you do not need to memorise chemical names. Reducing ultra-processed food removes the great majority of dietary emulsifiers automatically, and does several other useful things at once.

## Fructose

Named alongside ethanol and emulsifiers.[1] In practice this means sweetened drinks, high-fructose syrups, and large quantities of fruit juice rather than whole fruit.

**What to do:** cut liquid sugar first. It is the highest-dose, fastest-delivery form and the easiest to remove.

# What a head-to-head found among the popular gut diets

Elimination diets are where most people spend their effort, so it is worth knowing how they compare.

**Ten randomised trials covering 939 participants** compared four dietary interventions for irritable bowel syndrome. The low FODMAP diet showed relatively consistent short-term efficacy, with responder rates of 34% to 78%. **In pooled analysis there was no significant difference versus the other dietary interventions**, with a risk ratio of 1.04, a confidence interval of 0.91 to 1.19, and no heterogeneity.[3]

**Two things follow.** Low FODMAP genuinely works for a substantial minority. And it is not better than the alternatives, which matters because it is by far the most restrictive and the hardest to sustain.

**And it does not transfer to inflammatory bowel disease.** A review of five randomised trials found **no effect of a low FODMAP diet on disease activity** in Crohn's disease or ulcerative colitis.[4] It may help IBS-type symptoms in someone who also has IBD; it does not treat the IBD.

**How I would use that.** If you try low FODMAP, run it as a structured short-term trial with a planned reintroduction, not as a permanent way of eating. It restricts many of the fermentable fibers that feed the microbiome, and the trial data does not support paying that cost indefinitely.

# The foundation, which is ordinary and works

**Fiber**, named among the factors that improve barrier function.[1] Vegetables, legumes, whole grains if tolerated, fruit, nuts, and seeds. Increase gradually, because a fast jump causes the bloating people then blame on the food itself.

**Protein sufficient for repair**, which also supplies the tryptophan and cysteine named in the same review.[1] Eggs, fish, poultry, meat, legumes, dairy if tolerated.

**Vitamin A** from liver, eggs, dairy, and the orange and dark green vegetables. **Vitamin D** is difficult to obtain from food in useful amounts and is worth testing.

**Fermented foods** if you tolerate them. Yogurt, kefir, sauerkraut, kimchi. Reasonable, cheap, and pleasant, and their evidence is best treated as promising rather than settled.

**Minimal ultra-processed food**, which is how you remove emulsifiers, most added fructose, and a great deal else without reading a single label.

# Running a fair experiment

**Rule out first.** Celiac and inflammatory bowel disease, per the lab guide. An elimination diet run over an undiagnosed celiac wastes months and destroys the test.

**Then remove, one at a time.**

1. **Liquid sugar and fructose**, two weeks. Fastest to show anything.
2. **Alcohol**, four weeks off, not reduced.
3. **NSAIDs**, if your use is regular, in consultation with whoever prescribes for you.
4. **Ultra-processed food**, four weeks, which handles emulsifiers.

**Only then consider an elimination diet**, and only as a structured trial with a reintroduction phase built in from the start.

**Write down your baseline first.** Gut symptoms fluctuate more than most, and without a record you will be comparing today against an inaccurate memory.

**Track it:** rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=gut-diet

# Realistic expectations

The four stressors are documented in human work, and removing them is free. That is the strongest thing this guide offers.

Low FODMAP helps 34% to 78% of people with IBS short term, and is no better than the alternatives.[3] It does not treat IBD.[4]

The foundation pattern is ordinary nutrition with reasonable support for barrier function.[1] It is not a repair protocol and no diet has been shown to seal an intestinal barrier on a timeline.

**What I would not do** is take on three simultaneous restrictions on the strength of a permeability panel whose limitations are covered in the lab guide.

# When to seek clinical review

**Urgently:** blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain.

**Promptly:** a change in bowel habit lasting more than a few weeks, particularly over age 45.

**Before eliminating gluten:** celiac testing, while still eating gluten.

**Before a restrictive diet:** if you are underweight, have a history of disordered eating, are pregnant, or are managing diabetes. Stacked eliminations carry real nutritional risk.

# Working with Root Health

**Book a free Root Discovery Session:** rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=gut-diet

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# How you eat, and the two inputs that are not food

Digestion runs on the parasympathetic nervous system, so a meal eaten in a rush is processed differently from the same meal eaten sitting down. This is not a mood point, it is a mechanical one, and it costs nothing.

- Five slow breaths before the first bite, longer out than in.
- Sit down, no screens, no driving.
- Chew until the texture is gone.
- Stop at satisfied rather than full.

**And the two inputs that no diet fixes.** Acute stress produced a measurable increase in intestinal macromolecular permeability in healthy women in an experimental study, an effect not seen in men in the same experiment, building on earlier work showing chronic psychosocial stress induces jejunal barrier dysfunction (PMID 22625665). Sleep sits on the same gut-brain axis, with a 2026 review describing the bidirectional relationship while noting the mechanistic work is not finished (PMID 42092445).

If you change the food and nothing else, and stress and sleep are your actual barrier load, you will do this again next year.

# References


1. Camilleri M, Vella A. Human Intestinal Barrier: Effects of Stressors, Diet, Prebiotics, and Probiotics. *Clin Transl Gastroenterol.* 2021. PMID 33492118.

2. Effect of Five Dietary Emulsifiers on Inflammation, Permeability, and the Gut Microbiome: A Placebo-controlled Randomized Trial. *Clin Gastroenterol Hepatol.* 2026. PMID 40816342.

3. Characteristics and clinical applicability of four dietary interventions for irritable bowel syndrome. *Clin Nutr.* 2026. PMID 42160924.

4. Effects of a Low FODMAP Diet in Inflammatory Bowel Disease and Patient Experience. *J Hum Nutr Diet.* 2025. PMID 40765115.

5. Alonso C, Guilarte M, Vicario M, et al. Acute experimental stress evokes a differential gender-determined increase in human intestinal macromolecular permeability. *Neurogastroenterol Motil.* 2012;24(8):740-746. PMID 22625665

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**Seek care now** for blood in the stool, black tarry stool, unintended weight loss, persistent vomiting, or fever with abdominal pain. Those are never a supplement problem.

**Before anything in this guide:** the lab guide in this kit lists the diagnoses that get missed when symptoms are attributed to the barrier. Celiac disease and inflammatory bowel disease both have real treatments and both are worth excluding before you spend money here.

# Why nutrient evidence looks thinner than it is

**A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back.** You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

**First, "no evidence" has three different meanings, and this guide tries to always tell you which one applies.** *Tested and failed*: a real trial ran and the answer was no. *Never properly tested*: nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won*: real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

**Second, the asymmetry earns nutrients humility, not automatic credit.** When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

# Stress and sleep, which are barrier inputs rather than lifestyle advice

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- Movement most days at a sustainable intensity.

None of that costs anything, none of it interacts with anything, and leaving it out is why some people run three gut protocols and stay where they started.

# Start with the finding that should reorder your spending

**Glutamine is the flagship gut-repair supplement, and a meta-analysis found no overall effect on intestinal permeability.**

A systematic review and meta-analysis of clinical trials on glutamine and gut permeability in adults found that **glutamine supplementation did not significantly affect intestinal permeability**, with a weighted mean difference of 0.00 and a confidence interval of 0.04 to 0.03.[1]

The abstract does report a positive subgroup at higher doses. I am not building a recommendation on it, because that abstract states the threshold as over 30 g per day in one sentence and over 30 mg per day in the next, and reports a confidence interval that does not contain its own point estimate. When a result is reported that inconsistently, the honest move is to note it and not lean on it.

**And in inflammatory bowel disease specifically**, a systematic review found glutamine supplementation had **no effect on disease course, intestinal permeability, intestinal morphology, disease activity, symptoms, biochemical parameters, oxidative stress, or inflammation markers.**[2] That is a comprehensive list of things it did not change.

**The counterpoint, stated fairly.** A review of human barrier function describes glutamine and zinc as among the substances that enhance the barrier and can reverse the effect of stressors.[3] That is a narrative review drawing on mechanistic and animal work alongside human data, and it sits in tension with the clinical-trial meta-analysis. Where a narrative review and a meta-analysis of trials disagree, I weight the trials.

**What I would take from this.** Glutamine is not dangerous and it is not the foundation of gut repair it is marketed as. If you are spending meaningfully on it, that money is better placed on the modifiable stressors in the diet guide.

# Tier 1: The one category with real condition-specific evidence

# Probiotics, with the emphasis on specificity

An umbrella review of probiotics, prebiotics, and synbiotics concluded that these agents offer **promising but selective clinical benefits depending on the condition and population**, are generally safe, and that their application **should be condition specific**.[4]

That last phrase is the entire recommendation. “Take a probiotic for gut health” is not a plan. “Take this strain for this condition” is, and only some conditions have that evidence.

**Where the evidence is reasonable:** irritable bowel syndrome has the largest body of work. In pediatric functional abdominal pain disorders, a systematic review found probiotics give **modest symptom relief** and should be considered a safe adjunctive therapy rather than a stand-alone treatment, in line with 2025 ESPGHAN and NASPGHAN guidelines, while noting that larger standardized trials are still needed to define strain, dose, and duration.[5]

**Read “modest” as written.** It is a real effect and it is not transformative.

**Prebiotics, for contrast.** The barrier review describes effects of prebiotics on barrier function as **modest**, while noting more documented effects for probiotics, especially combination agents, across in vitro, animal, and human work.[3]

**Practically:** pick a product with named strains at stated counts, matched to your actual condition rather than to the word “gut,” give it four to eight weeks, and stop if nothing changes. A probiotic that has done nothing for three months is a subscription.

# Zinc, if you are low

Named in the barrier review alongside glutamine as a substance that enhances barrier function and can reverse stressor effects.[3] Unlike glutamine, zinc deficiency is a real and testable state with consequences beyond the gut.

**Functional target I use at Root Health:** 70 to 110 ug/dL plasma.

**Cautions:** long-term zinc drives copper deficiency, which causes its own problems including a neuropathy that mimics B12 deficiency. Do not take it open-endedly without a reason. Separate from antibiotics and from thyroid medication.

## Vitamins A and D, tryptophan, cysteine, and fiber

The same review lists these among dietary factors that improve the barrier.[3] Four of the five are food, and the diet guide in this kit covers them there, which is where they belong. Vitamin D is worth testing rather than guessing; the functional target is 40 to 80 ng/mL.

# Tier 2: Ask your practitioner first

**No products and no purchase links here.**

**Anything at all, if you have inflammatory bowel disease.** IBD is a treated disease with real medications, and the glutamine result above is a specific warning against assuming supplements substitute. *Ask:* "Does anything I am taking interact with my IBD medication, and is there anything you would want me to avoid?"

**High-dose zinc, long term.** Because of copper. *Ask:* "Should we check copper if I stay on this?"

**Anything, if you are on immunosuppressive therapy.** Live probiotic organisms in an immunosuppressed person are a genuine clinical question rather than a formality.

**If your symptoms have changed recently, or you are over 45 with a new change in bowel habit.** That is a workup, not a supplement conversation.

# Curcumin, with the trial that is usually oversold

The ebook names bioavailable curcumin in the barrier repair phase, so the honest evidence belongs here rather than in a footnote.

**What the research shows.** A randomised, double blind controlled trial assigned 206 patients with functional dyspepsia to curcumin, omeprazole, or both, for 28 days with follow-up to 56 days. All three groups improved significantly on dyspepsia scores, with no significant difference between them, no synergistic effect, and no serious adverse events (PMID 37696679).

**The caveat that decides how much weight to give it.** There was **no placebo arm.** Three active treatments, everyone improved, nobody differed. Functional dyspepsia has a large placebo response and a fluctuating natural course, so that design cannot separate “both treatments work” from “both are tracking placebo response and regression to the mean.” Also: 206 enrolled and 151 completed, and the paper carries a published correction from 2025 (PMID 38866470), which is a correction rather than a retraction.

**And the population caveat, which matters here.** That trial was in functional dyspepsia, not in intestinal permeability. There is no trial showing curcumin repairs a human gut barrier. Its inclusion rests on the anti-inflammatory mechanism described in the ebook plus this adjacent-condition trial, not on barrier outcome data.

**Typical studied range.** 250 mg four times daily, 2 g per day, in the trial above.

**Formulation is the part people get wrong.** Standard curcumin is poorly absorbed, which is why trials use enhanced delivery formats such as a phytosome or a piperine-paired preparation. Buying plain turmeric powder and expecting trial results is the common mistake.

**Cautions.** Increases bleeding risk with anticoagulants and antiplatelet drugs, stop before surgery. Can cause reflux and stomach upset. Interacts with several drugs through the same liver enzymes. Not established in pregnancy.

# What I deliberately left out, and why

**Most of the gut repair category.** Collagen, bone broth, aloe, slippery elm, deglycyrrhizinated licorice, and the various proprietary barrier blends. Some may help. I did not find human trial evidence that they change permeability or outcomes, and I am not going to imply otherwise.

**Anything sold as sealing or healing the gut lining on a stated timeline.** No supplement has demonstrated that.

**Proprietary blends with undisclosed amounts.** You cannot compare them to a studied dose or tell a clinician what you are taking.

**Things I use in practice that are not in here.** A guide sold to someone I have never examined has no history, no medication list, and no way to course-correct. The bar for a document is higher than the bar for a conversation. Where the evidence does not support recommending something at a distance, it stays out, and that is exactly why you can trust what did make it in.

# Running a fair trial

**Rule out first.** Celiac and IBD, per the lab guide. Everything else is guessing until those are excluded.

**Remove before you add.** The diet guide covers the stressors with actual human evidence: alcohol, NSAIDs, fructose, and emulsifiers. Removing a documented cause beats adding a speculative fix, and it is free.

**One at a time, four to eight weeks**, with a written baseline.

**Track it:** rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

# Where to find quality versions of these

Everything above was written without reference to any product line.

**Third-party testing first.** NSF Certified for Sport and USP Verified mean an independent lab confirmed the contents. **For probiotics specifically, look for named strains with counts guaranteed through the expiry date**, not at manufacture, since that is where cheap products fail.

**us.fullscript.com/welcome/drseay**

**Disclosure:** Root Health earns a portion of sales through that link. Nothing here was included, excluded, ranked, or dosed based on what it earns. The best-selling supplement in this entire category, glutamine, is the one this guide opens by telling you did not beat placebo.

# When to seek clinical review

**Urgently:** blood in stool, black tarry stool, unintended weight loss, persistent vomiting, fever with abdominal pain.

**Promptly:** a change in bowel habit lasting more than a few weeks, particularly over age 45.

**Before starting:** if you take immunosuppressive medication, or have IBD.

# Working with Root Health

**Book a free Root Discovery Session:** rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

**Get in touch:** rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=gut-supplement

# References


1. A systematic review and meta-analysis of clinical trials on the effects of glutamine supplementation on gut permeability in adults. *Amino Acids.* 2024. PMID 39397201.

2. Effects of glutamine supplementation on inflammatory bowel disease: A systematic review. *Clin Nutr ESPEN.* 2021. PMID 33745622.

3. Camilleri M, Vella A. Human Intestinal Barrier: Effects of Stressors, Diet, Prebiotics, and Probiotics. *Clin Transl Gastroenterol.* 2021. PMID 33492118. *Narrative review drawing on mechanistic, animal, and human data.*

4. Efficacy and Safety of Probiotics, Prebiotics, and Synbiotics: An Umbrella Review. *Nutr Rev.* 2026. PMID 42289829.

5. Probiotics in pediatric functional abdominal pain disorders: a systematic review. *Eur J Pediatr.* 2026. PMID 42477222.

6. Alonso C, Guilarte M, Vicario M, et al. Acute experimental stress evokes a differential gender-determined increase in human intestinal macromolecular permeability. *Neurogastroenterol Motil.* 2012;24(8):740-746. PMID 22625665

7. Andersen ML, Alvarenga TA, Tufik S. Cognition in the time of dysbiosis: Sleep disruption and the gut-brain dialogue. *Prog Neuropsychopharmacol Biol Psychiatry.* 2026;147:111730. PMID 42092445

8. Kongkam P, Khongkha W, Lopimpisuth C, et al. Curcumin and proton pump inhibitors for functional dyspepsia: a randomised, double blind controlled trial. *BMJ Evid Based Med.* 2023;28(6):399-406. PMID 37696679

9. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev.* 2017;2(2):MR000033. PMID 28207928


Functional target ranges transcribed from the Root Health blood chemistry range.



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